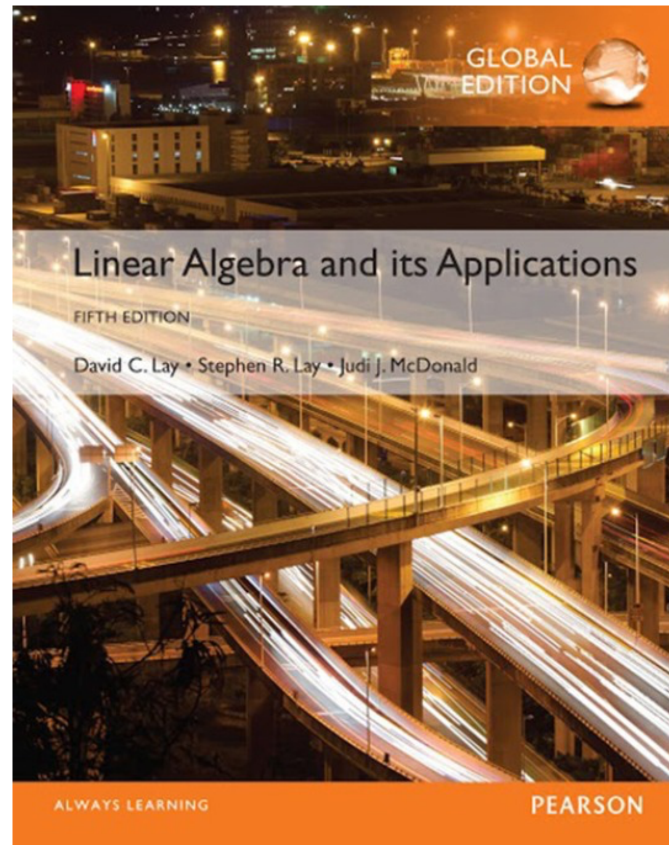
4 Vector Spaces

4.1

VECTOR SPACES AND SUBSPACES



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VECTOR SPACES AND SUBSPACES

- **Definition:** A **vector space** is a nonempty set V of objects, called *vectors*, on which are defined two operations, called *addition and multiplication by scalars* (real numbers), subject to the ten axioms (or rules) listed below. The axioms must hold for all vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ in V and for all scalars c and d.
  1. The sum of $\mathbf{u}$ and $\mathbf{v}$, denoted by $\mathbf{u}+\mathbf{v}$, is in V.
  2. $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$.
  3. $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$.
  4. There is a zero vector $\mathbf{0}$ in V such that

     $$\mathbf{u}+\mathbf{0}=\mathbf{u}.$$

VECTOR SPACES AND SUBSPACES

5. For each **u** in V, there is a vector $-\mathrm{u}$ in V such that $\mathrm{u}+(-\mathrm{u})=0$.
6. The scalar multiple of **u** by c, denoted by c**u**, is in V.
7. $c(\mathrm{u}+\mathrm{v})=c\mathrm{u}+c\mathrm{v}$.
8. $(c+d)\mathrm{u}=c\mathrm{u}+d\mathrm{u}$.
9. $c(d\mathrm{u})=(cd)\mathrm{u}$.
10. $1\mathrm{u}=\mathrm{u}$.

- Using these axioms, we can show that the zero vector in Axiom 4 is unique, and the vector $-\mathrm{u}$, called the **negative** of **u**, in Axiom 5 is unique for each **u** in V.

VECTOR SPACES AND SUBSPACES

- For each **u** in V and scalar c,

$$0\mathbf{u} = \mathbf{0} \qquad (1)$$

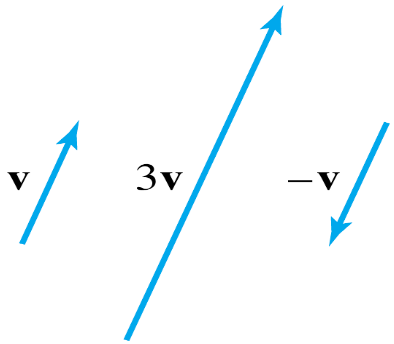
$$c\mathbf{0} = \mathbf{0} \qquad (2)$$

$$-\mathbf{u} = (-1)\mathbf{u} \qquad (3)$$

- **Example 2:** Let V be the set of all arrows (directed line segments) in three-dimensional space, with two arrows regarded as equal if they have the same length and point in the same direction. Define addition by the parallelogram rule, and for each **v** in V, define $c\mathbf{v}$ to be the arrow whose length is $|c|$ times the length of **v**, pointing in the same direction as **v** if $c \geq 0$ and otherwise pointing in the opposite direction.

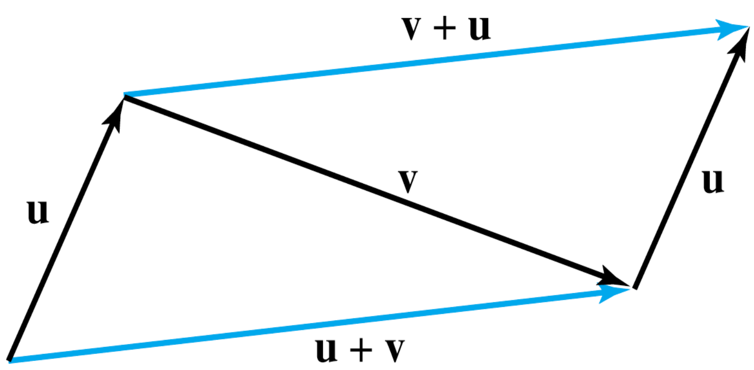
VECTOR SPACES AND SUBSPACES

- See the following figure below. Show that V is a vector space.

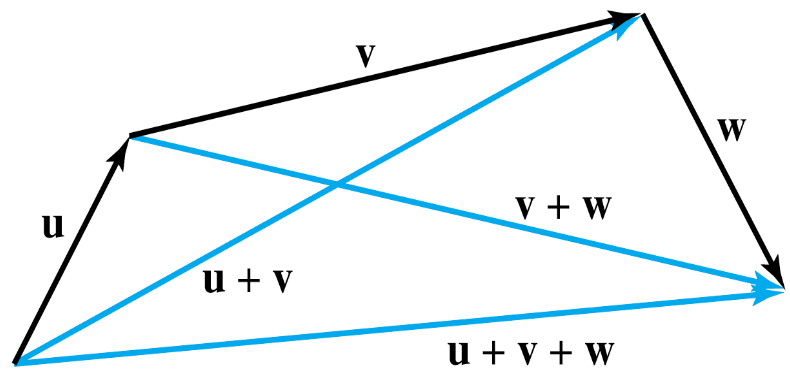


- **Solution:** The definition of V is geometric, using concepts of length and direction.
- No x y z-coordinate system is involved.
- An arrow of zero length is a single point and represents the zero vector.
- The negative of $\mathbf{v}$ is $(-1)\mathbf{v}$.
- So Axioms 1, 4, 5, 6, and 10 are evident. See the figures on the next slide.

SUBSPACES



$\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$.



$(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$.

- **Definition:** A **subspace** of a vector space V is a subset H of V that has three properties:
  a. The zero vector of V is in H.
  b. H is closed under vector addition. That is, for each $\mathbf{u}$ and $\mathbf{v}$ in H, the sum $\mathbf{u}+\mathbf{v}$ is in H.

SUBSPACES

c. H is closed under multiplication by scalars. That is, for each $\mathbf{u}$ in H and each scalar c, the vector $c\mathbf{u}$ is in H.

- Properties (a), (b), and (c) guarantee that a subspace H of V is itself a *vector space*, under the vector space operations already defined in V.
- Every subspace is a vector space.
- Conversely, every vector space is a subspace (of itself and possibly of other larger spaces).

A SUBSPACE SPANNED BY A SET

- The set consisting of only the zero vector in a vector space V is a subspace of V, called the **zero subspace** and written as $\{\mathbf{0}\}$.

- As the term **linear combination** refers to any sum of scalar multiples of vectors, and Span $\{\mathbf{v}_1,\ldots,\mathbf{v}_p\}$ denotes the set of all vectors that can be written as linear combinations of $\mathbf{v}_1,\ldots,\mathbf{v}_p$.

A SUBSPACE SPANNED BY A SET

- **Example 10:** Given $\mathbf{v}_1$ and $\mathbf{v}_2$ in a vector space V, let $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Show that H is a subspace of V.
- **Solution:** The zero vector is in H, since $\mathbf{0} = 0\mathbf{v}_1 + 0\mathbf{v}_2$.
- To show that H is closed under vector addition, take two arbitrary vectors in H, say,

$$\mathbf{u} = s_1\mathbf{v}_1 + s_2\mathbf{v}_2 \quad \text{and} \quad \mathbf{w} = t_1\mathbf{v}_1 + t_2\mathbf{v}_2.$$

- By Axioms 2, 3, and 8 for the vector space V,

$$\begin{aligned} \mathbf{u} + \mathbf{w} &= (s_1\mathbf{v}_1 + s_2\mathbf{v}_2) + (t_1\mathbf{v}_1 + t_2\mathbf{v}_2) \\ &= (s_1 + t_1)\mathbf{v}_1 + (s_2 + t_2)\mathbf{v}_2 \end{aligned}$$

A SUBSPACE SPANNED BY A SET

- So $\mathbf{u} + \mathbf{w}$ is in H.

- Furthermore, if c is any scalar, then by Axioms 7 and 9,
$$c\mathbf{u} = c(s_1\mathbf{v}_1 + s_2\mathbf{v}_2) = (cs_1)\mathbf{v}_1 + (cs_2)\mathbf{v}_2$$
which shows that $c\mathbf{u}$ is in H and H is closed under scalar multiplication.

- Thus H is a subspace of V.

A SUBSPACE SPANNED BY A SET

- **Theorem 1:** If $\mathbf{v}_1,\ldots,\mathbf{v}_p$ are in a vector space V, then Span $\{\mathbf{v}_1,\ldots,\mathbf{v}_p\}$ is a subspace of V.

- We call Span $\{\mathbf{v}_1,\ldots,\mathbf{v}_p\}$ **the subspace spanned** (or **generated**) by $\{\mathbf{v}_1,\ldots,\mathbf{v}_p\}$.

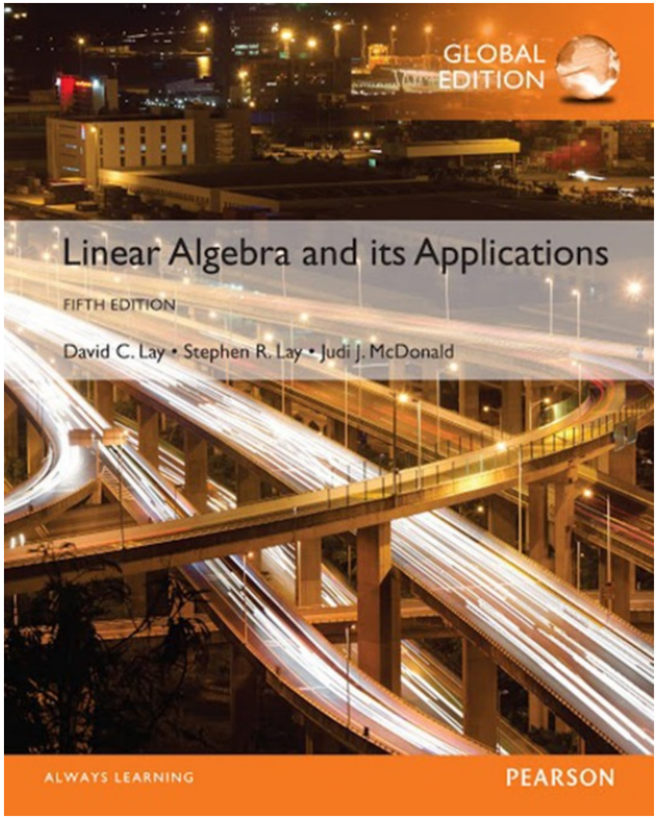
- Give any subspace H of V, a **spanning** (or **generating**) set for H is a set $\{\mathbf{v}_1,\ldots,\mathbf{v}_p\}$ in H such that

$$H = \text{Span}\{\text{v}_1,\ldots\text{v}_p\}.$$

4 Vector Spaces

4.2

NULL SPACES, COLUMN SPACES, AND LINEAR TRANSFORMATIONS



NULL SPACE OF A MATRIX

- **Definition:** The **null space** of an $m \times n$ matrix A, written as Nul$A$, is the set of all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$. In set notation,

$$\text{Nul } A = \{\mathbf{x} : \mathbf{x} \text{ is in } \mathbb{R}^n \text{ and } A\mathbf{x} = \mathbf{0}\}.$$

- **Theorem 2:** The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently, the set of all solutions to a system $A\mathbf{x} = \mathbf{0}$ of m homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.
- **Proof:** Nul$A$ is a subset of $\mathbb{R}^n$ because A has n columns.
- We need to show that Nul$A$ satisfies the three properties of a subspace.

NULL SPACE OF A MATRIX

- **0** is in Null A.
- Next, let **u** and **v** represent any two vectors in Nul A.
- Then

$$A\mathbf{u} = \mathbf{0} \text{ and } A\mathbf{v} = \mathbf{0}$$

- To show that $\mathbf{u} + \mathbf{v}$ is in Nul A, we must show that $A(\mathbf{u} + \mathbf{v}) = \mathbf{0}$.
- Using a property of matrix multiplication, compute

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}$$

- Thus $\mathbf{u} + \mathbf{v}$ is in Nul A, and Nul A is closed under vector addition.

NULL SPACE OF A MATRIX

- Finally, if c is any scalar, then

$$A(c\mathbf{u}) = c(A\mathbf{u}) = c(\mathbf{0}) = \mathbf{0}$$

which shows that $c\mathbf{u}$ is in Nul$A$.

- Thus Nul$A$ is a subspace of $\mathbb{R}^n$.

- **An Explicit Description of Nul*A***
- There is no obvious relation between vectors in Nul$A$ and the entries in A.
- We say that Nul$A$ is defined *implicitly*, because it is defined by a condition that must be checked.

NULL SPACE OF A MATRIX

- No explicit list or description of the elements in Nul A is given.
- *Solving* the equation $A\mathbf{x} = 0$ amounts to producing an *explicit* description of Nul A.

- **Example 3:** Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}.$$

NULL SPACE OF A MATRIX

- **Solution:** The first step is to find the general solution of $A\mathbf{x} = 0$ in terms of free variables.

- Row reduce the augmented matrix $\begin{bmatrix} A & 0 \end{bmatrix}$ to *reduce* echelon form in order to write the basic variables in terms of the free variables:

$$\begin{bmatrix} 1 & -2 & 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \qquad \begin{aligned} x_1 - 2x_2 - x_4 + 3x_5 &= 0 \\ x_3 + 2x_4 - 2x_5 &= 0 \\ 0 &= 0 \end{aligned}$$

NULL SPACE OF A MATRIX

- The general solution is $x_1 = 2x_2 + x_4 - 3x_5$, $x_3 = -2x_4 + 2x_5$, with x_2, x_4, and x_5 free.
- Next, decompose the vector giving the general solution into a linear combination of *vectors where the weights are the free variables*. That is,

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \underset{\substack{\uparrow \\ \mathbf{u}}}{\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}} + x_4 \underset{\substack{\uparrow \\ \mathbf{v}}}{\begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}} + x_5 \underset{\substack{\uparrow \\ \mathbf{w}}}{\begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}}$$

NULL SPACE OF A MATRIX

$$= x_2\mathbf{u} + x_4\mathbf{v} + x_5\mathbf{w} \qquad (3)$$

- Every linear combination of **u**, **v**, and **w** is an element of Nul A.
- Thus {**u**, **v**, **w**} is a spanning set for Nul A.

1. The spanning set produced by the method in Example (3) is automatically linearly independent because the free variables are the weights on the spanning vectors.
2. When Nul A contains nonzero vectors, the number of vectors in the spanning set for Nul A equals the number of free variables in the equation $A\mathbf{x} = 0$.

COLUMN SPACE OF A MATRIX

- **Definition:** The column space of an $m \times n$ matrix A, written as Col A, is the set of all linear combinations of the columns of A. If $A = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_n \end{bmatrix}$, then

$$\text{Col } A = \text{Span}\{\mathbf{a}_1, \ldots, \mathbf{a}_n\}.$$

- **Theorem 3:** The column space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^m$.
- A typical vector in Col A can be written as $A\mathbf{x}$ for some $\mathbf{x}$ because the notation $A\mathbf{x}$ stands for a linear combination of the columns of A. That is,

$$\text{Col } A = \{\mathbf{b} : \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \text{ in } \mathbb{R}^n\}.$$

COLUMN SPACE OF A MATRIX

- The notation $A\mathbf{x}$ for vectors in Col A also shows that Col A is the *range* of the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$.
- The column space of an $m \times n$ matrix A is all of $\mathbb{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for each $\mathbf{b}$ in $\mathbb{R}^m$.
- **Example 7:** Let $A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}$.

COLUMN SPACE OF A MATRIX

a. Determine if **u** is in Nul A. Could **u** be in Col A?
b. Determine if **v** is in Col A. Could **v** be in Nul A?

- **Solution:**

a. An explicit description of Nul A is not needed here. Simply compute the product $A\mathbf{u}$.

$$A\mathrm{u} = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

COLUMN SPACE OF A MATRIX

- **u** is *not* a solution of $A\mathrm{x} = 0$, so **u** is not in Nul$A$.
- Also, with four entries, **u** could not possibly be in Col A, since Col A is a subspace of $\mathbb{R}^3$.

b. Reduce $\begin{bmatrix} A & \mathrm{v} \end{bmatrix}$ to an echelon form.

$$\begin{bmatrix} A & \mathrm{v} \end{bmatrix} = \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{bmatrix} \sim \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ 0 & 1 & -5 & -4 & -2 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}$$

- The equation $A\mathrm{x} = \mathrm{v}$ is consistent, so **v** is in Col A.
- With only three entries, **v** could not be in Nul$A$.

CONTRAST BETWEEN NUL *A* AND COL *A* FOR AN $m \times n$ MATRIX *A*

Nul$A$	Col A
1. Nul A is a subspace of $\mathbb{R}^n$.	1. Col A is a subspace of $\mathbb{R}^m$.
2. Nul$A$ is implicitly defined; *i.e.*, you are given only a condition $(A\mathrm{x} = 0)$ that vectors in Nul$A$ must satisfy.	2. Col A is explicitly defined; *i.e.*, you are told how to build vectors in Col A.

CONTRAST BETWEEN NUL A AND COL A FOR AN $m \times n$ MATRIX A

3. It takes time to find vectors in Nul A. Row operations on $[A \quad 0]$ are required.	3. It is easy to find vectors in Col A. The columns of a are displayed; others are formed from them.
4. There is no obvious relation between Nul A and the entries in A.	4. There is an obvious relation between Col A and the entries in A, since each column of A is in Col A.

CONTRAST BETWEEN NUL *A* AND COL *A* FOR AN $m \times n$ MATRIX *A*

5. A typical vector **v** in Nul *A* has the property that $A\mathbf{v} = 0$.	5. A typical vector **v** in Col *A* has the property that the equation $A\mathbf{x} = \mathbf{v}$ is consistent.
6. Given a specific vector **v**, it is easy to tell if **v** is in Nul *A*. Just compare $A\mathbf{v}$.	6. Given a specific vector **v**, it may take time to tell if v is in Col *A*. Row operations on $\begin{bmatrix} A & \mathbf{v} \end{bmatrix}$ are required.

CONTRAST BETWEEN NUL *A* AND COL *A* FOR AN $m \times n$ MATRIX *A*

7. Nul $A = \{0\}$ if and only if the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.	7. Col $A = \mathbb{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution for every **b** in $\mathbb{R}^m$.
8. Nul $A = \{0\}$ if and only if the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.	8. Col $A = \mathbb{R}^m$ if and only if the linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^m$.

KERNEL AND RANGE OF A LINEAR TRANSFORMATION

- Subspaces of vector spaces other than $\mathbb{R}^n$ are often described in terms of a linear transformation instead of a matrix.
- **Definition:** A **linear transformation** T from a vector space V into a vector space W is a rule that assigns to each vector **x** in V a unique vector $T(\mathbf{x})$ in W, such that
  i. $T(\mathrm{u}+\mathrm{v}) = T(\mathrm{u}) + T(\mathrm{v})$ for all **u**, **v** in V, and
  ii. $T(c\mathrm{u}) = cT(\mathrm{u})$ for all **u** in V and all scalars c.

KERNEL AND RANGE OF A LINEAR TRANSFORMATION

- The **kernel** (or **null space**) of such a T is the set of all $\mathbf{u}$ in V such that $T(\mathrm{u}) = 0$ (the zero vector in W).

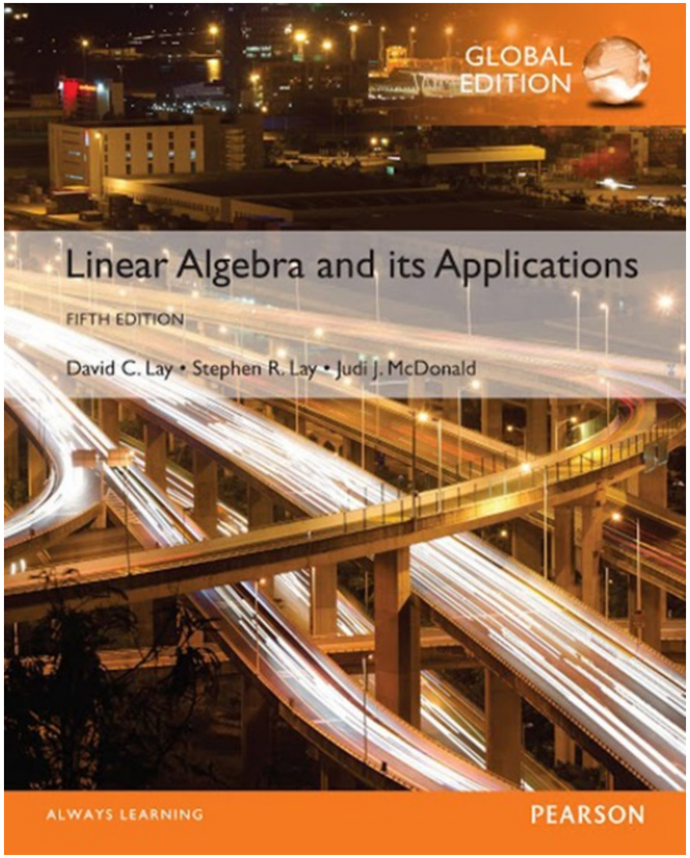
- The **range** of T is the set of all vectors in W of the form $T(\mathbf{x})$ for some $\mathbf{x}$ in V.

- The kernel of T is a subspace of V.

- The range of T is a subspace of W.

4 Vector Spaces

4.3 LINEARLY INDEPENDENT SETS; BASES



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LINEAR INDEPENDENT SETS; BASES

- An indexed set of vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ in V is said to be **linearly independent** if the vector equation

$$c_1\mathrm{v}_1 + c_2\mathrm{v}_2 + \ldots + c_p\mathrm{v}_p = 0 \qquad (1)$$

  has *only* the trivial solution, $c_1 = 0, \ldots, c_p = 0$.

- The set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is said to be **linearly dependent** if (1) has a nontrivial solution, *i.e.*, if there are some weights, $c_1, \ldots, c_p$, *not all zero*, such that (1) holds.

- In such a case, (1) is called a **linear dependence relation** among $\mathbf{v}_1, \ldots, \mathbf{v}_p$.

LINEAR INDEPENDENT SETS; BASES

- **Theorem 4:** An indexed set $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ of two or more vectors, with $\mathrm{v}_1 \neq 0$, is linearly dependent if and only if some $\mathbf{v}_j$ (with $j > 1$) is a linear combination of the preceding vectors, $\mathrm{v}_1, \ldots, \mathrm{v}_{j-1}$.

- **Definition:** Let H be a subspace of a vector space V. An indexed set of vectors $\mathcal{B} = \{\mathrm{b}_1, \ldots, \mathrm{b}_p\}$ in V is a basis for H if
  - (i) $\mathcal{B}$ is a linearly independent set, and
  - (ii) The subspace spanned by $\mathcal{B}$ coincides with H; that is, $H = \mathrm{Span}\{\mathrm{b}_1, \ldots, \mathrm{b}_p\}$

LINEAR INDEPENDENT SETS; BASES

- The definition of a basis applies to the case when $H = V$, because any vector space is a subspace of itself.

- Thus a basis of V is a linearly independent set that spans V.

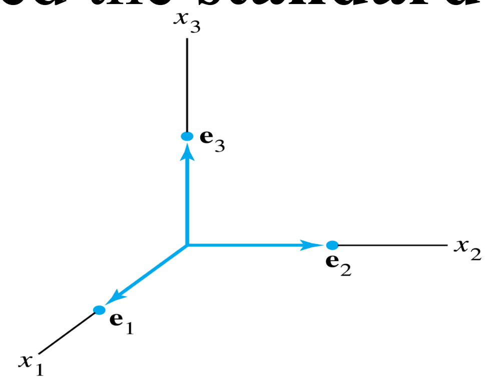
- When $H \neq V$, condition (ii) includes the requirement that each of the vectors $\mathbf{b}_1, \ldots, \mathbf{b}_p$ must belong to H, because Span $\{\mathbf{b}_1, \ldots, \mathbf{b}_p\}$ contains $\mathbf{b}_1, \ldots, \mathbf{b}_p$.

STANDARD BASIS

- Let $\mathbf{e}_1, \ldots, \mathbf{e}_n$ be the columns of the $n \times n$ matrix, I_n.
- That is,

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \ldots, e_n = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}$$

- The set $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is called the **standard basis** for $\mathbb{R}^n$. See the following figure.



The standard basis for $\mathbb{R}^3$.

THE SPANNING SET THEOREM

- **Theorem 5:** Let $S = \{v_1, \ldots, v_p\}$ be a set in V, and let $H = \text{Span}\{v_1, \ldots, v_p\}$.
  a. If one of the vectors in S—say, $\mathbf{v}_k$—is a linear combination of the remaining vectors in S, then the set formed from S by removing $\mathbf{v}_k$ still spans H.
  b. If $H \neq \{0\}$, some subset of S is a basis for H.
- **Proof:**
  a. By rearranging the list of vectors in S, if necessary, we may suppose that $\mathbf{v}_p$ is a linear combination of $v_1, \ldots, v_{p-1}$—say,

THE SPANNING SET THEOREM

$$\mathbf{v}_p = a_1\mathbf{v}_1 + \ldots + a_{p-1}\mathbf{v}_{p-1} \qquad (3)$$

- Given any $\mathbf{x}$ in H, we may write
$$\mathbf{x} = c_1\mathbf{v}_1 + \ldots + c_{p-1}\mathbf{v}_{p-1} + c_p\mathbf{v}_p \qquad (4)$$
for suitable scalars $c_1, \ldots, c_p$.

- Substituting the expression for $\mathbf{v}_p$ from (3) into (4), it is easy to see that $\mathbf{x}$ is a linear combination of $\mathbf{v}_1, \ldots \mathbf{v}_{p-1}$.

- Thus $\{\mathbf{v}_1, \ldots, \mathbf{v}_{p-1}\}$ spans H, because $\mathbf{x}$ was an arbitrary element of H.

THE SPANNING SET THEOREM

b. If the original spanning set S is linearly independent, then it is already a basis for H.

- Otherwise, one of the vectors in S depends on the others and can be deleted, by part (a).
- So long as there are two or more vectors in the spanning set, we can repeat this process until the spanning set is linearly independent and hence is a basis for H.
- If the spanning set is eventually reduced to one vector, that vector will be nonzero (and hence linearly independent) because $H \neq \{0\}$.

THE SPANNING SET THEOREM

- **Example 7:** Let $v_1 = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 6 \\ 16 \\ -5 \end{bmatrix}$

  and $H = \text{Span}\{v_1, v_2, v_3\}$.
  Note that $v_3 = 5v_1 + 3v_2$, and show that $\text{Span}\{v_1, v_2, v_3\} = \text{Span}\{v_1, v_2\}$. Then find a basis for the subspace H.

- **Solution:** Every vector in Span $\{\mathbf{v}_1, \mathbf{v}_2\}$ belongs to H because

$$c_1 v_1 + c_2 v_2 = c_1 v_1 + c_2 v_2 + 0 v_3$$

THE SPANNING SET THEOREM

- Now let **x** be any vector in H—say,

$$\mathrm{x} = c_1\mathrm{v}_1 + c_2\mathrm{v}_2 + c_3\mathrm{v}_3 .$$

- Since $\mathrm{v}_3 = 5\mathrm{v}_1 + 3\mathrm{v}_2$, we may substitute

$$\mathrm{x} = c_1\mathrm{v}_1 + c_2\mathrm{v}_2 + c_3(5\mathrm{v}_1 + 3\mathrm{v}_2)$$

$$= (c_1 + 5c_3)\mathrm{v}_1 + (c_2 + 3c_3)\mathrm{v}_2$$

- Thus **x** is in Span $\{\mathbf{v}_1, \mathbf{v}_2\}$, so every vector in H already belongs to Span $\{\mathbf{v}_1, \mathbf{v}_2\}$.
- We conclude that H and Span $\{\mathbf{v}_1, \mathbf{v}_2\}$ are actually the same sets of vectors.
- It follows that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a basis of H since $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent.

BASIS FOR COL B

- **Example 8:** Find a basis for Col B, where

$$B = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_5 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 0 & 2 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- **Solution:** Each nonpivot column of B is a linear combination of the pivot columns.
- In fact, $\mathbf{b}_2 = 4\mathbf{b}_1$ and $\mathbf{b}_4 = 2\mathbf{b}_1 - \mathbf{b}_3$.
- By the Spanning Set Theorem, we may discard $\mathbf{b}_2$ and $\mathbf{b}_4$, and $\{\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_5\}$ will still span Col B.

BASIS FOR COL B

- Let

$$S = \{\mathbf{b}_1, \mathbf{b}_3, \mathbf{b}_5\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$$

- Since $\mathbf{b}_1 \neq 0$ and no vector in S is a linear combination of the vectors that precede it, S is linearly independent. (Theorem 4).

- Thus S is a basis for Col B.

BASES FOR NUL A AND COL A

- **Theorem 6:** The pivot columns of a matrix A form a basis for Col A.

- **Proof:** Let B be the reduced echelon form of A.
- The set of pivot columns of B is linearly independent, for no vector in the set is a linear combination of the vectors that precede it.
- Since A is row equivalent to B, the pivot columns of A are linearly independent as well, because any linear dependence relation among the columns of A corresponds to a linear dependence relation among the columns of B.

BASES FOR NUL *A* AND COL *A*

- For this reason, every nonpivot column of A is a linear combination of the pivot columns of A.
- Thus the nonpivot columns of a may be discarded from the spanning set for Col A, by the Spanning Set Theorem.
- This leaves the pivot columns of A as a basis for Col A.

- **Warning:** The pivot columns of a matrix A are evident when A has been reduced only to echelon form.
- But, be careful to use the pivot columns of A itself for the basis of Col A.

BASES FOR NUL *A* AND COL *A*

- Row operations can change the column space of a matrix.
- The columns of an echelon form B of A are often not in the column space of A.

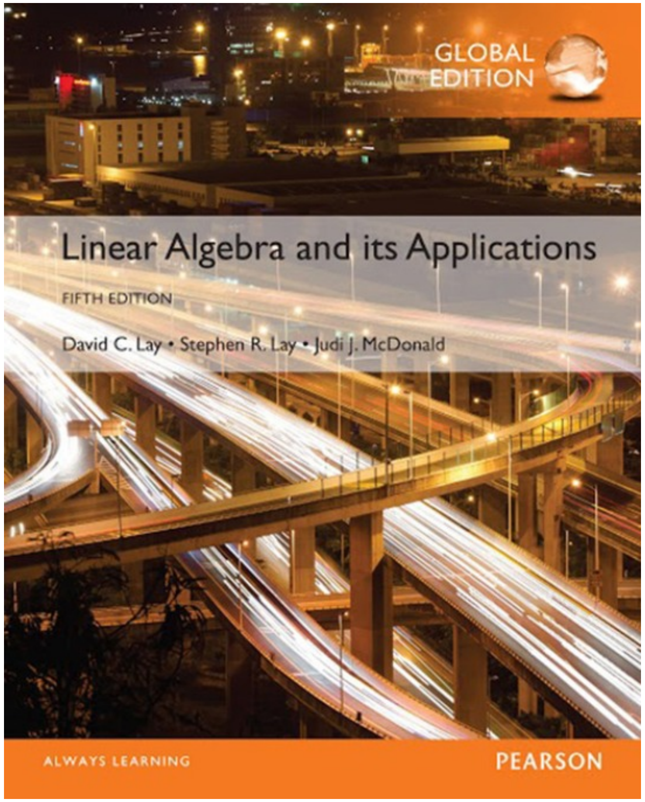
- **Two Views of a Basis**
- When the Spanning Set Theorem is used, the deletion of vectors from a spanning set must stop when the set becomes linearly independent.
- If an additional vector is deleted, it will not be a linear combination of the remaining vectors, and hence the smaller set will no longer span V.

TWO VIEWS OF A BASIS

- Thus a basis is a spanning set that is as small as possible.
- A basis is also a linearly independent set that is as large as possible.
- If S is a basis for V, and if S is enlarged by one vector—say, $\mathbf{w}$—from V, then the new set cannot be linearly independent, because S spans V, and $\mathbf{w}$ is therefore a linear combination of the elements in S.

4 Vector Spaces

4.4 COORDINATE SYSTEMS



THE UNIQUE REPRESENTATION THEOREM

- **Theorem 7:** Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for vector space V. Then for each **x** in V, there exists a unique set of scalars $c_1, \ldots, c_n$ such that

$$\mathbf{x} = c_1\mathbf{b}_1 + \ldots + c_n\mathbf{b}_n \qquad (1)$$

- **Proof:** Since $\mathcal{B}$ spans V, there exist scalars such that (1) holds.
- Suppose **x** also has the representation

$$\mathbf{x} = d_1\mathbf{b}_1 + \ldots + d_n\mathbf{b}_n$$

for scalars $d_1, \ldots, d_n$.

THE UNIQUE REPRESENTATION THEOREM

- Then, subtracting, we have
$$0 = \mathbf{x} - \mathbf{x} = (c_1 - d_1)\mathbf{b}_1 + \ldots + (c_n - d_n)\mathbf{b}_n \qquad (2)$$

- Since B is linearly independent, the weights in (2) must all be zero. That is, $c_j = d_j$ for $1 \le j \le n$.

- **Definition:** Suppose $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ is a basis for V and **x** is in V. **The coordinates of x relative to the basis B** (or the **B-coordinate of x**) are the weights c_1, …, c_n such that $\mathbf{x} = c_1\mathbf{b}_1 + \ldots + c_n\mathbf{b}_n$

THE UNIQUE REPRESENTATION THEOREM

- If $c_1, \ldots, c_n$ are the **B**-coordinates of **x**, then the vector in $\mathbb{R}^n$

$$[\mathrm{x}]_{\mathrm{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

  is the **coordinate vector of x (relative to B)**, or the **B-coordinate vector of x**.

- The mapping $\mathrm{x} \mapsto [\mathrm{x}]_{\mathrm{B}}$ is the **coordinate mapping (determined by B)**.

COORDINATES IN $\mathbb{R}^n$

- When a basis B for $\mathbb{R}^n$ is fixed, the B-coordinate vector of a specified **x** is easily found, as in the example below.
- **Example 1:** Let $b_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, b_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}, x = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$, and

B $= \{b_1, b_2\}$. Find the coordinate vector $[\mathbf{x}]_B$ of **x** relative to B.

- **Solution:** The B-coordinate c_1, c_2 of **x** satisfy

$$c_1 \underbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{b_1} + c_2 \underbrace{\begin{bmatrix} -1 \\ 1 \end{bmatrix}}_{b_2} = \underbrace{\begin{bmatrix} 4 \\ 5 \end{bmatrix}}_{\mathbf{x}}$$

COORDINATES IN $\mathbb{R}^n$

or

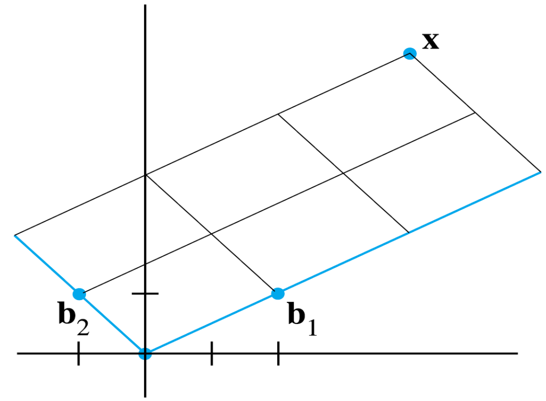
$$\begin{array}{cc} \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 5 \end{bmatrix} \\ \mathbf{b}_1 \quad \mathbf{b}_2 \qquad\qquad\quad \mathbf{x} \end{array} \tag{3}$$

- This equation can be solved by row operations on an augmented matrix or by using the inverse of the matrix on the left.
- In any case, the solution is $c_1 = 3, c_2 = 2$.
- Thus $\mathbf{x} = 3\mathbf{b}_1 + 2\mathbf{b}_2$ and

$$[\mathbf{x}]_B = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

COORDINATES IN $\mathbb{R}^n$

- See the following figure.



The $\mathcal{B}$-coordinate vector of **x** is (3, 2).

- The matrix in (3) changes the B-coordinates of a vector **x** into the standard coordinates for **x**.
- An analogous change of coordinates can be carried out in $\mathbb{R}^n$ for a basis $\mathrm{B} = \{\mathrm{b}_1, \ldots, \mathrm{b}_n\}$.
- Let $P_{\mathrm{B}} = \begin{bmatrix} \mathrm{b}_1 & \mathrm{b}_2 & \cdots & \mathrm{b}_n \end{bmatrix}$

COORDINATES IN $\mathbb{R}^n$

- Then the vector equation

$$\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \ldots + c_n\mathbf{b}_n$$

is equivalent to

(4) $$\boxed{\mathbf{x} = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}}$$

- $P_{\mathcal{B}}$ is called the **change-of-coordinates matrix** from $\mathcal{B}$ to the standard basis in $\mathbb{R}^n$.

- Left-multiplication by $P_{\mathcal{B}}$ transforms the coordinate vector $[\mathbf{x}]_{\mathcal{B}}$ into $\mathbf{x}$.

- Since the columns of $P_{\mathcal{B}}$ form a basis for $\mathbb{R}^n$, $P_{\mathcal{B}}$ is invertible (by the Invertible Matrix Theorem).

COORDINATES IN $\mathbb{R}^n$

- Left-multiplication by $P_{\mathcal{B}}^{-1}$ converts **x** into its $\mathcal{B}$-coordinate vector:

$$P_{\mathcal{B}}^{-1}\mathbf{x} = [\mathbf{x}]_{\mathcal{B}}$$

- The correspondence $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$, produced by $P_{\mathcal{B}}^{-1}$, is the coordinate mapping.

- Since $P_{\mathcal{B}}^{-1}$ is an invertible matrix, the coordinate mapping is a one-to-one linear transformation from $\mathbb{R}^n$ onto $\mathbb{R}^n$, by the Invertible Matrix Theorem.

THE COORDINATE MAPPING

- **Theorem 8:** Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for a vector space V. Then the coordinate mapping $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

- **Proof:** Take two typical vectors in V, say,

$$\mathbf{u} = c_1\mathbf{b}_1 + \ldots + c_n\mathbf{b}_n$$

$$\mathbf{w} = d_1\mathbf{b}_1 + \ldots + d_n\mathbf{b}_n$$

- Then, using vector operations,

$$\mathbf{u} + \mathbf{v} = (c_1 + d_1)\mathbf{b}_1 + \ldots + (c_n + d_n)\mathbf{b}_n$$

THE COORDINATE MAPPING

- It follows that

$$[\mathbf{u}+\mathbf{w}]_{\mathrm{B}} = \begin{bmatrix} c_1 + d_1 \\ \vdots \\ c_n + d_n \end{bmatrix} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} + \begin{bmatrix} d_1 \\ \vdots \\ d_n \end{bmatrix} = [\mathbf{u}]_{\mathrm{B}} + [\mathbf{w}]_{\mathrm{B}}$$

- So the coordinate mapping preserves addition.

- If r is any scalar, then

$$r\mathbf{u} = r(c_1\mathbf{b}_1 + \ldots + c_n\mathbf{b}_n) = (rc_1)\mathbf{b}_1 + \ldots + (rc_n)\mathbf{b}_n$$

THE COORDINATE MAPPING

- So

$$[r\mathbf{u}]_B = \begin{bmatrix} rc_1 \\ \vdots \\ rc_n \end{bmatrix} = r\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} = r[\mathbf{u}]_B$$

- Thus the coordinate mapping also preserves scalar multiplication and hence is a linear transformation.
- The linearity of the coordinate mapping extends to linear combinations.
- If $\mathbf{u}_1, \ldots, \mathbf{u}_p$ are in V and if $c_1, \ldots, cp$ are scalars, then

$$\left[c_1\mathbf{u}_1 + \ldots + c_p\mathbf{u}_p\right]_B = c_1\left[\mathbf{u}_1\right]_B + \ldots + c_p\left[\mathbf{u}_p\right]_B \quad (5)$$

THE COORDINATE MAPPING

- In words, (5) says that the B-coordinate vector of a linear combination of $\mathbf{u}_1, \ldots, \mathbf{u}_p$ is the *same* linear combination of their coordinate vectors.
- The coordinate mapping in Theorem 8 is an important example of an *isomorphism* from V onto $\mathbb{R}^n$.
- In general, a one-to-one linear transformation from a vector space V onto a vector space W is called an **isomorphism** from V onto W.
- The notation and terminology for V and W may differ, but the two spaces are indistinguishable as vector spaces.

THE COORDINATE MAPPING

- *Every vector space calculation in V is accurately reproduced in W, and vice versa.*
- In particular, any real vector space with a basis of n vectors is indistinguishable from $\mathbb{R}^n$.
- **Example 7:** Let $\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \ \mathbf{x} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix},$

  and $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$. Then $\mathcal{B}$ is a basis for $H = \text{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$. Determine if $\mathbf{x}$ is in H, and if it is, find the coordinate vector of $\mathbf{x}$ relative to $\mathcal{B}$.

THE COORDINATE MAPPING

- **Solution:** If $\mathbf{x}$ is in H, then the following vector equation is consistent:

$$c_1 \begin{bmatrix} 3 \\ 6 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 12 \\ 7 \end{bmatrix}$$

- The scalars c_1 and c_2, if they exist, are the B-coordinates of $\mathbf{x}$.

THE COORDINATE MAPPING

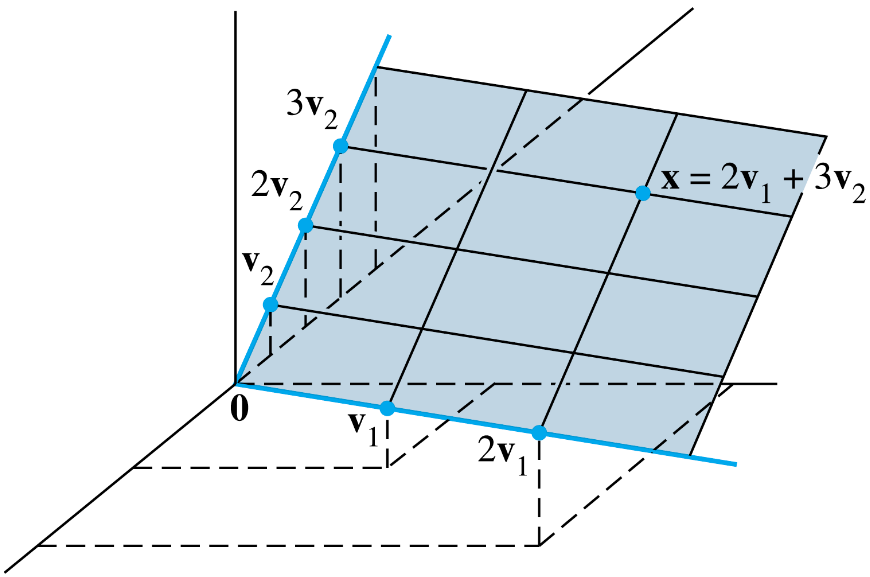
- Using row operations, we obtain

$$\begin{bmatrix} 3 & -1 & 3 \\ 6 & 0 & 12 \\ 2 & 1 & 7 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$$

- Thus $c_1 = 2$, $c_2 = 3$ and $[\mathbf{x}]_B = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$.

THE COORDINATE MAPPING

- The coordinate system on H determined by $\mathcal{B}$ is shown in the following figure.

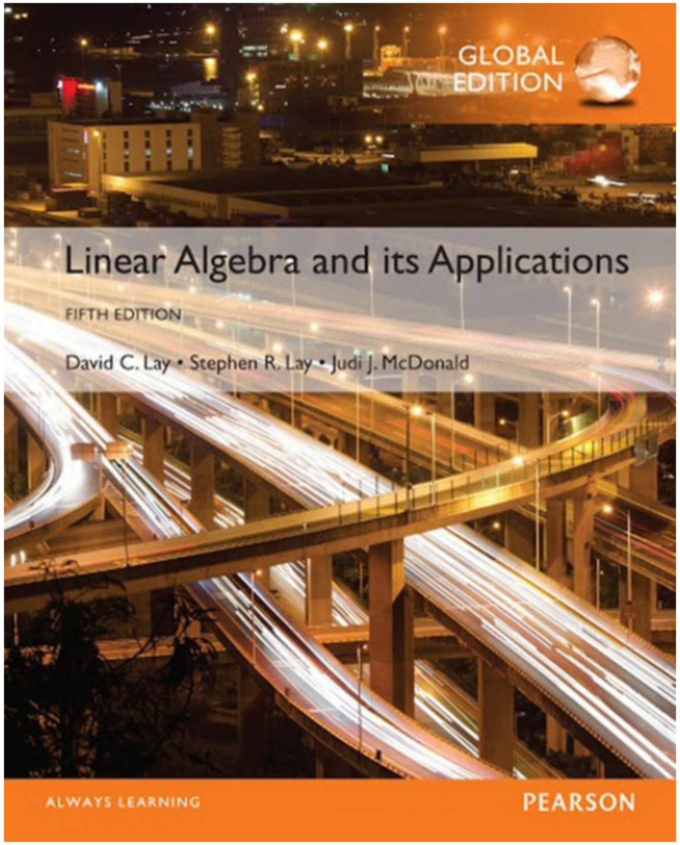


A coordinate system on a plane H in $\mathbb{R}^3$.

4 Vector Spaces

4.5

THE DIMENSION OF A VECTOR SPACE



DIMENSION OF A VECTOR SPACE

- **Theorem 9:** If a vector space V has a basis $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$, then any set in V containing more than n vectors must be linearly dependent.

- **Proof:** Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ be a set in V with more than n vectors.

- The coordinate vectors $[\mathbf{u}_1]_{\mathcal{B}}, \ldots, [\mathbf{u}_p]_{\mathcal{B}}$ form a linearly dependent set in $\mathbb{R}^n$, because there are more vectors (p) than entries (n) in each vector.

DIMENSION OF A VECTOR SPACE

- So there exist scalars $c_1, \ldots, c_p$, not all zero, such that

$$c_1[\mathrm{u}_1]_{\mathcal{B}} + \ldots + c_p[\mathrm{u}_p]_{\mathcal{B}} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

The zero vector in $\mathbb{R}^n$

- Since the coordinate mapping is a linear transformation,

$$[c_1\mathrm{u}_1 + \ldots + c_p\mathrm{u}_p]_{\mathcal{B}} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$$

DIMENSION OF A VECTOR SPACE

- The zero vector on the right displays the n weights needed to build the vector $c_1\mathrm{u}_1 + ... + c_p\mathrm{u}_p$ from the basis vectors in B.
- That is, $c_1\mathrm{u}_1 + ... + c_p\mathrm{u}_p = 0 \cdot \mathrm{b}_1 + ... + 0 \cdot \mathrm{b}_n = 0$.
- Since the c_i are not all zero, $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is linearly dependent.
- Theorem 9 implies that if a vector space V has a basis $\mathrm{B} = \{\mathrm{b}_1, ..., \mathrm{b}_n\}$, then each linearly independent set in V has no more than n vectors.

DIMENSION OF A VECTOR SPACE

- **Theorem 10:** If a vector space V has a basis of n vectors, then every basis of V must consist of exactly n vectors.

- **Proof:** Let B_1 be a basis of n vectors and B_2 be any other basis (of V).
- Since B_1 is a basis and B_2 is linearly independent, B_2 has no more than n vectors, by Theorem 9.
- Also, since B_2 is a basis and B_1 is linearly independent, B_2 has at least n vectors.
- Thus B_2 consists of exactly n vectors.

DIMENSION OF A VECTOR SPACE

- **Definition:** If V is spanned by a finite set, then V is said to be **finite-dimensional**, and the **dimension** of V, written as dim V, is the number of vectors in a basis for V. The dimension of the zero vector space $\{\mathbf{0}\}$ is defined to be zero. If V is not spanned by a finite set, then V is said to be **infinite-dimensional**.
- **Example 3:** Find the dimension of the subspace

$$H = \left\{ \begin{bmatrix} a-3b+6c \\ 5a+4d \\ b-2c-d \\ 5d \end{bmatrix} : a,b,c,d \text{ in } \mathbb{R} \right\}$$

DIMENSION OF A VECTOR SPACE

- H is the set of all linear combinations of the vectors

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 6 \\ 0 \\ -2 \\ 0 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 0 \\ 4 \\ -1 \\ 5 \end{bmatrix}$$

- Clearly, $\mathbf{v}_1 \neq 0$, $\mathbf{v}_2$ is not a multiple of $\mathbf{v}_1$, but $\mathbf{v}_3$ is a multiple of $\mathbf{v}_2$.
- By the Spanning Set Theorem, we may discard $\mathbf{v}_3$ and still have a set that spans H.

SUBSPACES OF A FINITE-DIMENSIONAL SPACE

- Finally, $\mathbf{v}_4$ is not a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.
- So $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is linearly independent and hence is a basis for H.
- Thus $\dim H = 3$.

- **Theorem 11:** Let H be a subspace of a finite-dimensional vector space V. Any linearly independent set in H can be expanded, if necessary, to a basis for H. Also, H is finite-dimensional and

$$\dim H \leq \dim V$$

SUBSPACES OF A FINITE-DIMENSIONAL SPACE

- **Proof:** If $H = \{0\}$, then certainly $\dim H = 0 \leq \dim V$.
- Otherwise, let $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ be any linearly independent set in H.
- If S spans H, then S is a basis for H.
- Otherwise, there is some $\mathbf{u}_{k+1}$ in H that is not in Span S.

SUBSPACES OF A FINITE-DIMENSIONAL SPACE

- But then $\{\mathrm{u}_1, \ldots, \mathrm{u}_k, \mathrm{u}_{k+1}\}$ will be linearly independent, because no vector in the set can be a linear combination of vectors that precede it (by Theorem 4).

- So long as the new set does not span H, we can continue this process of expanding S to a larger linearly independent set in H.

- But the number of vectors in a linearly independent expansion of S can never exceed the dimension of V, by Theorem 9.

THE BASIS THEOREM

- So eventually the expansion of S will span H and hence will be a basis for H, and $\dim H \leq \dim V$.

- **Theorem 12:** Let V be a p-dimensional vector space, $p \geq 1$. Any linearly independent set of exactly p elements in V is automatically a basis for V. Any set of exactly p elements that spans V is automatically a basis for V.

- **Proof:** By Theorem 11, a linearly independent set S of p elements can be extended to a basis for V.

THE BASIS THEOREM

- But that basis must contain exactly p elements, since $\dim V = p$.
- So S must already be a basis for V.
- Now suppose that S has p elements and spans V.
- Since V is nonzero, the Spanning Set Theorem implies that a subset S' of S is a basis of V.
- Since $\dim V = p$, S' must contain p vectors.
- Hence $S = S'$.

THE DIMENSIONS OF NUL *A* AND COL *A*

- Let A be an $m \times n$ matrix, and suppose the equation $A\mathrm{x} = 0$ has k free variables.

- A spanning set for Nul A will produce exactly k linearly independent vectors—say, $\mathrm{u}_1, \ldots, \mathrm{u}_k$—one for each free variable.

- So $\{\mathrm{u}_1, \ldots, \mathrm{u}_k\}$ is a basis for Nul A, and the number of free variables determines the size of the basis.

DIMENSIONS OF NUL *A* AND COL *A*

- Thus, the dimension of Nul A is the number of free variables in the equation $A\mathrm{x}=0$, and the dimension of Col A is the number of pivot columns in A.

- **Example 5:** Find the dimensions of the null space and the column space of

$$A=\begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

DIMENSIONS OF NUL *A* AND COL *A*

- **Solution:** Row reduce the augmented matrix $\begin{bmatrix} A & 0 \end{bmatrix}$ to echelon form:

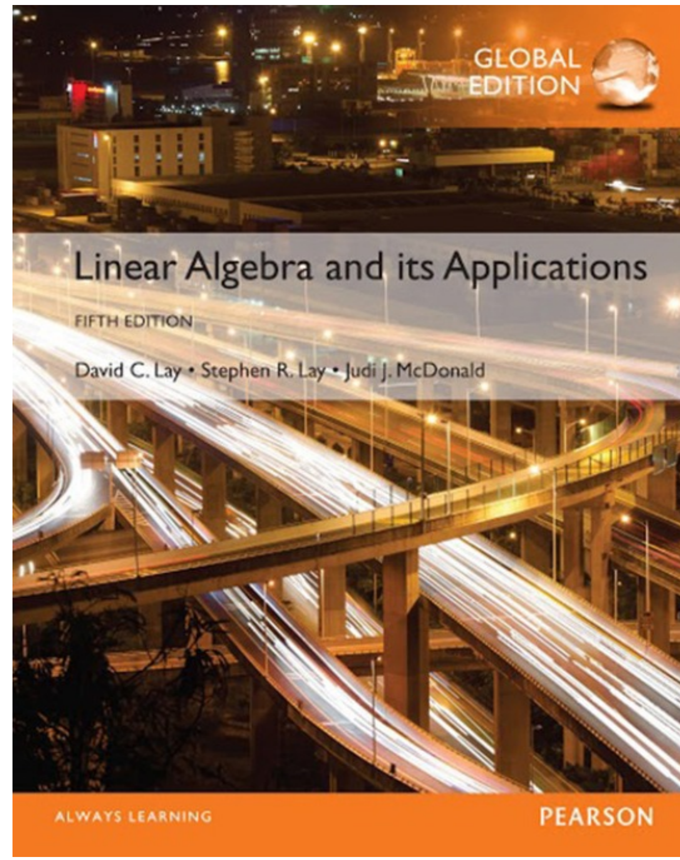
$$\begin{bmatrix} 1 & -2 & 2 & 3 & -1 & 0 \\ 0 & 0 & 1 & 2 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- There are three free variable—x_2, x_4 and x_5.
- Hence the dimension of Nul A is 3.
- Also dim Col $A = 2$ because A has two pivot columns.

4 Vector Spaces

4.6

RANK



THE ROW SPACE

- If A is an $m \times n$ matrix, each row of A has n entries and thus can be identified with a vector in $\mathbb{R}^n$.
- The set of all linear combinations of the row vectors is called the **row space** of A and is denoted by Row A.
- Each row has n entries, so Row A is a subspace of $\mathbb{R}^n$.
- Since the rows of A are identified with the columns of A^T, we could also write Col A^T in place of Row A.

THE ROW SPACE

- **Theorem 13:** If two matrices A and B are row equivalent, then their row spaces are the same. If B is in echelon form, the nonzero rows of B form a basis for the row space of A as well as for that of B.

- **Proof:** If B is obtained from A by row operations, the rows of B are linear combinations of the rows of A.

- It follows that any linear combination of the rows of B is automatically a linear combination of the rows of A.

THE ROW SPACE

- Thus the row space of B is contained in the row space of A.
- Since row operations are reversible, the same argument shows that the row space of A is a subset of the row space of B.
- So the two row spaces are the same.

THE ROW SPACE

- If B is in echelon form, its nonzero rows are linearly independent because no nonzero row is a linear combination of the nonzero rows below it. (Apply Theorem 4 to the nonzero rows of B in reverse order, with the first row last).

- Thus the nonzero rows of B form a basis of the (common) row space of B and A.

THE ROW SPACE

- **Example 2:** Find bases for the row space, the column space, and the null space of the matrix

$$\begin{bmatrix} -2 & -5 & 8 & 0 & -17 \\ 1 & 3 & -5 & 1 & 5 \\ 3 & 11 & -19 & 7 & 1 \\ 1 & 7 & -13 & 5 & -3 \end{bmatrix}$$

- **Solution:** To find bases for the row space and the column space, row reduce A to an echelon form:

THE ROW SPACE

$$A \sim B = \begin{bmatrix} 1 & 3 & -5 & 1 & 5 \\ 0 & 1 & -2 & 2 & -7 \\ 0 & 0 & 0 & -4 & 20 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- By Theorem 13, the first three rows of B form a basis for the row space of A (as well as for the row space of B).

- Thus

  Basis for Row $A : \{(1,3,-5,1,5),(0,1,-2,2,-7),(0,0,0,-4,20)\}$

THE ROW SPACE

- For the column space, observe from B that the pivots are in columns 1, 2, and 4.
- Hence columns 1, 2, and 4 of A (not B) form a basis for Col A:

$$\text{Basis for Col } A: \left\{ \begin{bmatrix} -2 \\ 1 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 3 \\ 11 \\ 7 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 7 \\ 5 \end{bmatrix} \right\}$$

- Notice that any echelon form of A provides (in its nonzero rows) a basis for Row A and also identifies the pivot columns of A for Col A.

THE ROW SPACE

- However, for Nul A, we need the *reduced echelon form*.

- Further row operations on B yield

$$A \sim B \sim C = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -2 & 0 & 3 \\ 0 & 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

THE ROW SPACE

- The equation $A\mathrm{x} = 0$ is equivalent to $C\mathrm{x} = 0$, that is,

$$\begin{aligned} x_1 + x_3 + x_5 &= 0 \\ x_2 - 2x_3 + 3x_5 &= 0 \\ x_4 - 5x_5 &= 0 \end{aligned}$$

- So $x_1 = -x_3 - x_5$, $x_2 = 2x_3 - 3x_5$, $x_4 = 5x_5$, with x_3 and x_5 free variables.

THE ROW SPACE

- The calculations show that

$$\text{Basis for Nul } A: \left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -3 \\ 0 \\ 5 \\ 1 \end{bmatrix} \right\}$$

- Observe that, unlike the basis for Col A, the bases for Row A and Nul A have no simple connection with the entries in A itself.

THE RANK THEOREM

- **Definition:** The **rank** of A is the dimension of the column space of A.
- Since Row A is the same as Col A^T, the dimension of the row space of A is the rank of A^T.
- The dimension of the null space is sometimes called the **nullity** of A.
- **Theorem 14:** The dimensions of the column space and the row space of an $m \times n$ matrix A are equal. This common dimension, the rank of A, also equals the number of pivot positions in A and satisfies the equation

$$\text{rank } A + \dim \text{Nul } A = n$$

THE RANK THEOREM

- **Proof:** By Theorem 6, rank A is the number of pivot columns in A.
- Equivalently, rank A is the number of pivot positions in an echelon form B of A.
- Since B has a nonzero row for each pivot, and since these rows form a basis for the row space of A, the rank of A is also the dimension of the row space.
- The dimension of Nul A equals the number of free variables in the equation $A\mathrm{x} = 0$.
- Expressed another way, the dimension of Nul A is the number of columns of A that are *not* pivot columns.

THE RANK THEOREM

- (It is the number of these columns, not the columns themselves, that is related to Nul A).

- Obviously,

$$\left\{\begin{array}{c}\text{number of}\\ \text{pivot columns}\end{array}\right\}+\left\{\begin{array}{c}\text{number of}\\ \text{nonpivot columns}\end{array}\right\}=\left\{\begin{array}{c}\text{number of}\\ \text{columns}\end{array}\right\}$$

- This proves the theorem.

THE RANK THEOREM

- **Example 3:**
  a. If A is a 7×9 matrix with a two-dimensional null space, what is the rank of A?
  b. Could a 6×9 matrix have a two-dimensional null space?
- **Solution:**
  a. Since A has 9 columns, $(\text{rank } A) + 2 = 9$, and hence rank $A = 7$.
  b. No. If a 6×9 matrix, call it B, has a two-dimensional null space, it would have to have rank 7, by the Rank Theorem.

THE INVERTIBLE MATRIX THEOREM (CONTINUED)

- But the columns of B are vectors in $\mathbb{R}^6$, and so the dimension of Col B cannot exceed 6; that is, rank B cannot exceed 6.

- **Theorem:** Let A be an $n \times n$ matrix. Then the following statements are each equivalent to the statement that A is an invertible matrix.
  m. The columns of A form a basis of $\mathbb{R}^n$.
  n. Col $A = \mathbb{R}^n$
  o. Dim Col $A = n$
  p. rank $A = n$

RANK AND THE INVERTIBLE MATRIX THEOREM

q. $\text{Nul } A = \{0\}$

r. $\text{Dim Nul } A = 0$

- **Proof:** Statement (m) is logically equivalent to statements (e) and (h) regarding linear independence and spanning.

- The other five statements are linked to the earlier ones of the theorem by the following chain of almost trivial implications:

$$(g) \Rightarrow (n) \Rightarrow (o) \Rightarrow (p) \Rightarrow (r) \Rightarrow (q) \Rightarrow (d)$$

RANK AND THE INVERTIBLE MATRIX THEOREM

- Statement (g), which says that the equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each **b** in $\mathbb{R}^n$, implies (n), because Col A is precisely the set of all **b** such that the equation $A\mathbf{x} = \mathbf{b}$ is consistent.

- The implications $(\text{n}) \Rightarrow (\text{o}) \Rightarrow (\text{p})$ follow from the definitions of dimension and rank.

- If the rank of A is n, the number of columns of A, then $\dim \text{Nul } A = 0$, by the Rank Theorem, and so $\text{Nul } A = \{0\}$.

RANK AND THE INVERTIBLE MATRIX THEOREM

- Thus $(p) \Rightarrow (r) \Rightarrow (q)$.

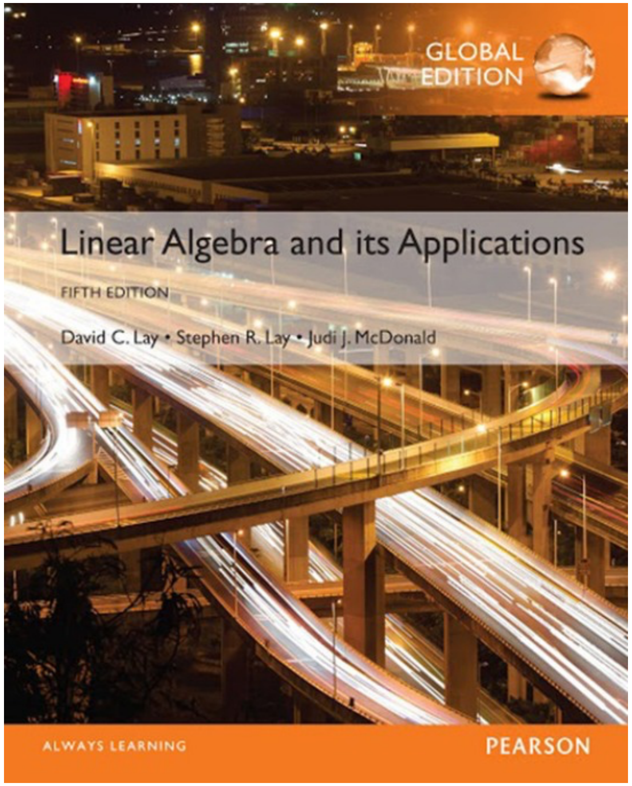
- Also, (q) implies that the equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, which is statement (d).

- Since statements (d) and (g) are already known to be equivalent to the statement that A is invertible, the proof is complete.

4 Determinants

4.7

CHANGE OF BASIS



CHANGE OF BASIS

- **Example 1** Consider two bases $\beta = \{b_1, b_2\}$ and $C = \{c_1, c_2\}$ for a vector space V, such that

$$b_1 = 4c_1 + c_2 \quad \text{and} \quad b_2 = -6c_1 + c_2 \qquad (1)$$

- Suppose

$$x = 3b_1 + b_2 \qquad (2)$$

- That is, suppose $[x]_\beta = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Find $[x]_C$.

CHANGE OF BASIS

- **Solution** Apply the coordinate mapping determined by C to **x** in (2). Since the coordinate mapping is a linear transformation,

$$\begin{aligned}[x]_C &= [3\mathrm{b}_1 + \mathrm{b}_2]_C \\ &= [3\mathrm{b}_1]_C + [\mathrm{b}_2]_C\end{aligned}$$

- We can write the vector equation as a matrix equation, using the vectors in the linear combination as the columns of a matrix:

$$[x]_C = [\ [\mathrm{b}_1]_C\ [\mathrm{b}_2]_C\ ]\begin{bmatrix} 3 \\ 1 \end{bmatrix} \tag{3}$$

CHANGE OF BASIS

- This formula gives $[x]_C$, once we know the columns of the matrix. From (1),

$$[\mathrm{b}_1]_C = \begin{bmatrix} 4 \\ 1 \end{bmatrix} \text{and } [\mathrm{b}_2]_C = \begin{bmatrix} -6 \\ 1 \end{bmatrix}$$

- Thus, (3) provides the solution:

$$[x]_C = \begin{bmatrix} 4 & -6 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

- The *C*-coordinates of **x** match those of the **x** in Fig. 1, as seen on the next slide.

CHANGE OF BASIS

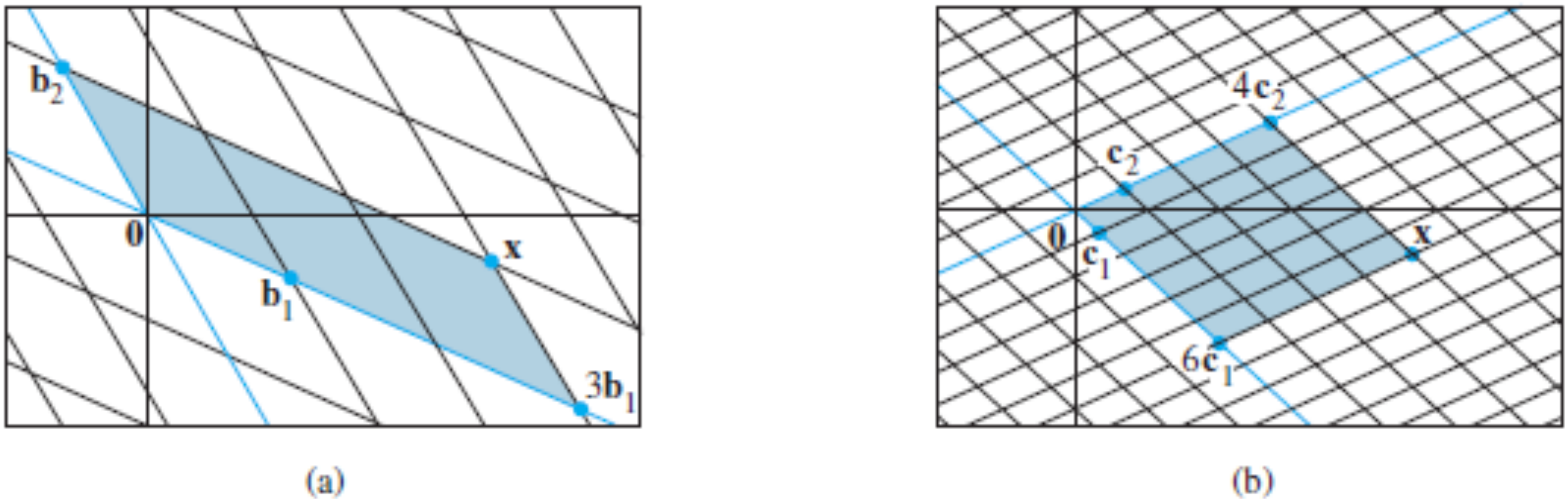


FIGURE 1 Two coordinate systems for the same vector space.

CHANGE OF BASIS

- **Theorem 15**: Let $\beta = \{b_1, \ldots, b_n\}$ and $C = \{c_1, \ldots, c_2\}$ for a vector space V. Then there is a unique n × n matrix $\underset{C \leftarrow \beta}{P}$ such that

$$[x]_C = \underset{C \leftarrow \beta}{P} [x]_\beta \tag{4}$$

- The columns of $\underset{C \leftarrow \beta}{P}$ are the C-coordinate vectors of the vectors in the basis β. That is,

$$\underset{C \leftarrow \beta}{P} = [\ [b_1]_C [b_2]_C \quad \ldots \quad [b_n]_C] \tag{5}$$

CHANGE OF BASIS

- The matrix $\underset{\mathcal{C}\leftarrow\mathcal{B}}{P}$ in Theorem 15 is called the **change-of-coordinates matrix from $\mathcal{B}$ to $\mathcal{C}$**. Multiplication by $\underset{\mathcal{C}\leftarrow\mathcal{B}}{P}$ converts $\mathcal{B}$-coordinates into $\mathcal{C}$-coordinates.
- Figure 2 below illustrates the change-of-coordinates equation (4).

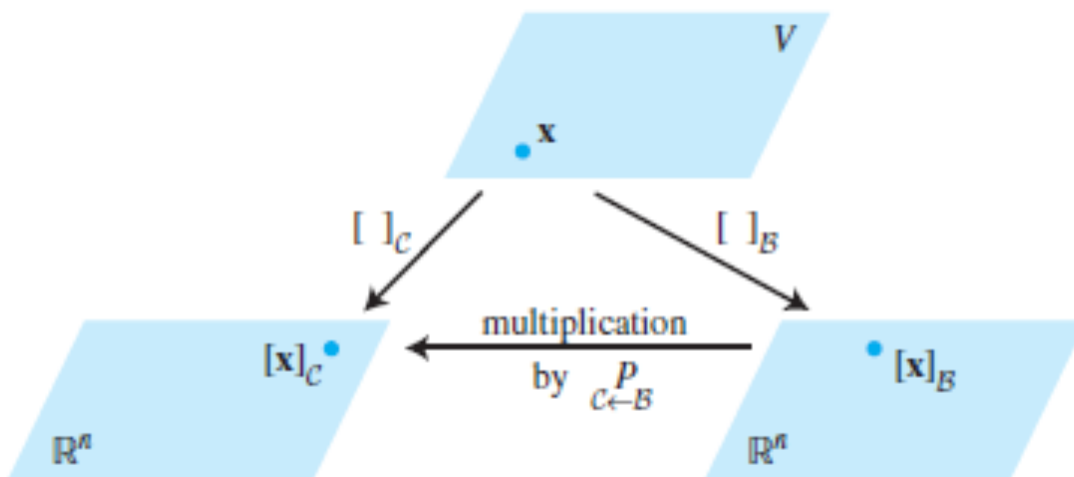


FIGURE 2 Two coordinate systems for V.

CHANGE OF BASIS

- The columns of $\underset{c \leftarrow \beta}{P}$ are linearly independent because they are the coordinate vectors of the linearly independent set β.

- Since $\underset{c \leftarrow \beta}{P}$ is square, it must be invertible, by the Invertible Matrix Theorem. Left-multiplying both sides of equation (4) by $(\underset{c \leftarrow \beta}{P})^{-1}$ yields

$$(\underset{c \leftarrow \beta}{P})^{-1} [x]_C = [x]_\beta$$

- Thus $(\underset{c \leftarrow \beta}{P})^{-1}$ is the matrix that converts C-coordinates into β–coordinates. That is,

$$(\underset{c \leftarrow \beta}{P})^{-1} = \underset{\beta \leftarrow C}{P} \qquad (6)$$

CHANGE OF BASIS IN $\mathbb{R}^n$

- If $\beta = \{b_1, \ldots, b_n\}$ and $\mathcal{E}$ is the standard basis $\{e_1, \ldots, e_n\}$ in $\mathbb{R}^n$, then $[b_1]_{\mathcal{E}} = b_1$, and likewise for the other vectors in β. In this case, $\underset{\mathcal{E} \leftarrow \beta}{P}$ is the same as the change-of-coordinates matrix P_β introduced in Section 4.4, namely,

$$P_\beta = [\, b_1 \quad b_2 \ldots b_n]$$

- To change coordinates between two nonstandard bases in $\mathbb{R}^n$, we need Theorem 15. The theorem shows that to solve the change-of-basis problem, we need the coordinate vectors of the old basis relative to the new basis.

CHANGE OF BASIS IN $\mathbb{R}^n$

- **Example 2** Let $\mathbf{b}_1 = \begin{bmatrix} -9 \\ 1 \end{bmatrix}$, $\mathbf{b}_2 = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$, $\mathbf{c}_1 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$, $\mathbf{c}_2 = \begin{bmatrix} 3 \\ -5 \end{bmatrix}$ and consider the bases for $\mathbb{R}^n$ given by $\beta = \{\mathbf{b}_1, \mathbf{b}_2\}$ and $C = \{\mathbf{c}_1, \mathbf{c}_2\}$. Find the change-of-coordinates matrix from β to C.

- **Solution** The matrix $\underset{C \leftarrow \beta}{P}$ involves the C-coordinate vectors of $\mathbf{b}_1$ and $\mathbf{b}_2$. Let $[\mathbf{b}_1]_C = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ and $[\mathbf{b}_2]_C = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$. Then, by definition,

$$[c_1 \quad c_2] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \mathbf{b}_1 \quad \text{and} \quad [c_1 \quad c_2] \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \mathbf{b}_2$$

CHANGE OF BASIS IN $\mathbb{R}^n$

- To solve both systems simultaneously, augment the coefficient matrix with b_1 and b_2, and row reduce:

$$[c_1 \quad c_2 \vdots b_1 \quad b_2] = \begin{bmatrix} 1 & 3 & \vdots & -9 & -5 \\ -4 & -5 & \vdots & 1 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \vdots & 6 & 4 \\ 0 & 1 & \vdots & -5 & -3 \end{bmatrix} \qquad (7)$$

- Thus

$$[b_1]_C = \begin{bmatrix} 6 \\ -5 \end{bmatrix} \text{and } [b_2]_C = \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

- The desired change-of-coordinates matrix is therefore

$$\underset{C \leftarrow \beta}{P} = [[b_1]_c \quad [b_2]_c] = \begin{bmatrix} 6 & 4 \\ -5 & -3 \end{bmatrix}$$